Ivan Morozov: Publication list

[1] E. Karantzoulis, A. Fabris, I. Morozov, K. Manukyan, M. Modica, N. Shafqat, S. Dastan, S. D. Mitri, and S. Krecic. "Elettra 2.0 - The new frontier of synchrotron radiation". English. In: *Proc. IPAC'26* (Deauville, France, May 17–22, 2026). IPAC'26 - 17th International Particle Accelerator Conference 17. presented at IPAC'26, Deauville, France, 2026, paper THP2039, to be published. JACoW Publishing, Geneva, Switzerland, May 2026, pp. 3515–3518. ISBN: 978-3-95450-252-3. URL: https://jacow.org/ipac2026/pdf/THP2039.pdf.

[2] I. Morozov and M. Giovannozzi. "Estimating the Stability Domain of Symplectic Maps: A Robust Method via Bounding Set of Unstable Initial Conditions". English. In: *Proc. IPAC'26* (Deauville, France, May 17–22, 2026). IPAC'26 - 17th International Particle Accelerator Conference 17. presented at IPAC'26, Deauville, France, 2026, paper WEP5028, to be published. JACoW Publishing, Geneva, Switzerland, May 2026, pp. 2627–2631. ISBN: 978-3-95450-252-3. URL: https://jacow.org/ipac2026/pdf/WEP5028.pdf.

[3] I. Morozov and M. Yuliya. "Applications of High-Dimensional Time-Delayed Embedding for Time Series Complexity Estimation". English. In: *Proc. IPAC'26* (Deauville, France, May 17–22, 2026). IPAC'26 - 17th International Particle Accelerator Conference 17. presented at IPAC'26, Deauville, France, 2026, paper WEP5027, to be published. JACoW Publishing, Geneva, Switzerland, May 2026, pp. 2622–2626. ISBN: 978-3-95450-252-3. URL: https://jacow.org/ipac2026/pdf/WEP5027.pdf.

[4] I. Morozov, B. Diviacco, E. Karantzoulis, K. Manukyan, S. Dastan, and S. Krecic. "Tracking-Based linear symplectic parameterization of insertion devices". English. In: *Proc. IPAC'26* (Deauville, France, May 17–22, 2026). IPAC'26 - 17th International Particle Accelerator Conference 17. presented at IPAC'26, Deauville, France, 2026, paper WEP5026, to be published. JACoW Publishing, Geneva, Switzerland, May 2026, pp. 2618–2621. ISBN: 978-3-95450-252-3. URL: https://jacow.org/ipac2026/pdf/WEP5026.pdf.

[5] T. Zolkin, S. Nagaitsev, I. Morozov, and S. Kladov. "Geometry of almost-conserved quantities in symplectic maps: Approximate invariants in nonlinear accelerator systems". In: *Phys. Rev. Accel. Beams* (Apr. 2026). DOI: 10.1103/zxgv-2xrf. URL: https://link.aps.org/doi/10.1103/zxgv-2xrf.

[6] T. Zolkin, S. Nagaitsev, I. Morozov, and S. Kladov. "Dynamics of McMillan Mappings III. Symmetric Map with Mixed Nonlinearity". In: *Nonlinear Dynamics* 114.9 (May 1, 2026), p. 635. ISSN: 1573-269X. DOI: 10.1007/s11071-026-12509-5. URL: https://doi.org/10.1007/s11071-026-12509-5.

[7] T. Zolkin, S. Nagaitsev, I. Morozov, and S. Kladov. "Geometry of almost-conserved quantities in symplectic maps". In: *Chaos, Solitons Fractals* 208 (2026), p. 118059. ISSN: 0960-0779. DOI: https://doi.org/10.1016/j.chaos.2026.118059. URL: https://www.sciencedirect.com/science/article/pii/S0960077926002006.

[8] V. Gauzshtein et al. "Rescattering effects in the reaction $\gamma d \to \pi^- pp$". In: *Scientific Reports* 16.1 (2026), p. 4003. ISSN: 2045-2322. DOI: 10.1038/s41598-025-34162-5. URL: https://doi.org/10.1038/s41598-025-34162-5.

[9] V. Anashin et al. "The branching fraction measurements of J/ decay into and final states". In: *Physics Letters B* (2026), p. 140152. ISSN: 0370-2693. DOI: https://doi.org/10.1016/j.physletb.2026.140152. URL: https://www.sciencedirect.com/science/article/pii/S0370269326000067.

[10] V. V. Anashin et al. "New measurement of D0 and D+ meson masses with the KEDR detector". In: *Journal of High Energy Physics* 2025.11 (Nov. 4, 2025), p. 1. ISSN: 1029-8479. DOI: 10.1007/JHEP11(2025)001. URL: https://doi.org/10.1007/JHEP11(2025)001.

[11] Y. Maltseva and I. Morozov. "Aperture limitation localization using beam position and beam loss monitor measurements". In: *Nuclear Instruments and Methods in Physics Research Section A: Accelerators, Spectrometers, Detectors and Associated Equipment* (2025), p. 170825. ISSN: 0168-9002. DOI: https://doi.org/10.1016/j.nima.2025.170825. URL: https://www.sciencedirect.com/science/article/pii/S0168900225006278.

[12] Y. Maltseva and I. Morozov. *Aperture limitation localization using beam position and beam loss monitor measurements.* 2025. arXiv: 2507.03333 [physics.acc-ph]. URL: https://arxiv.org/abs/2507.03333.

[13] P. Piminov et al. "Y(1S)-meson rest mass measurement on the VEPP-4M collider". English. In: *Proc. 16th International Particle Accelerator Conference* (Taipei, Taiwan). IPAC'25 - 16th International Particle Accelerator Conference 16. JACoW Publishing, Geneva, Switzerland, June 2025, pp. 568–570. ISBN: 978-3-95450-248-6. DOI: 10.18429/JACoW-IPAC25-MOPM115. URL: https://indico.jacow.org/event/81/contributions/7325.

[14] V. V. Anashin et al. "Measurement of the Masses of Neutral and Charged D Mesons with the KEDR Detector". In: *Physics of Particles and Nuclei* 56.3 (June 2025), pp. 802–808. ISSN: 1531-8559. DOI: 10.1134/S1063779624702320. URL: https://doi.org/10.1134/S1063779624702320.

[15] V. V. Anashin et al. *The branching fraction measurements of J/ψ decay into ρη and φη final states*. 2025. arXiv: 2506.08453 [hep-ex]. URL: https://arxiv.org/abs/2506.08453.

[16] V. V. Anashin et al. *New measurement of D0 and D+ meson masses with the KEDR detector*. 2025. arXiv: 2506.02421 [hep-ex]. URL: https://arxiv.org/abs/2506.02421.

[17] T. Zolkin, S. Nagaitsev, I. Morozov, and S. Kladov. *Geometry of Almost-Conserved Quantities in Symplectic Maps. Part III: Approximate Invariants in Nonlinear Accelerator Systems*. 2025. arXiv: 2505.07225 [nlin.CD]. URL: https://arxiv.org/abs/2505.07225.

[18] T. Zolkin, S. Nagaitsev, I. Morozov, and S. Kladov. *Geometry of Almost-Conserved Quantities in Symplectic Maps. Part II: Recovery of approximate invariant*. 2025. arXiv: 2505.07224 [nlin.CD]. URL: https://arxiv.org/abs/2505.07224.

[19] T. Zolkin, S. Nagaitsev, I. Morozov, and S. Kladov. *Geometry of Almost-Conserved Quantities in Symplectic Maps. Part I: Perturbation Theory*. 2025. arXiv: 2505.07223 [nlin.CD]. URL: https://arxiv.org/abs/2505.07223.

[20] T. Zolkin, S. Nagaitsev, I. Morozov, S. Kladov, and Y.-K. Kim. "Isochronous and period-doubling diagrams for symplectic maps of the plane". In: *Chaos, Solitons Fractals* 198 (2025), p. 116513. ISSN: 0960-0779. DOI: https://doi.org/10.1016/j.chaos.2025.116513. URL: https://www.sciencedirect.com/science/article/pii/S0960077925005260.

[21] T. Zolkin, S. Nagaitsev, and I. Morozov. "Dynamics of McMillan mappings I. McMillan multipoles". In: *Physica D: Nonlinear Phenomena* (2025), p. 134620. ISSN: 0167-2789. DOI: https://doi.org/10.1016/j.physd.2025.134620. URL: https://www.sciencedirect.com/science/article/pii/S0167278925000995.

[22] I. Morozov and Y. Maltseva. "Coupled twiss parameters estimation from turn-by-turn data". In: *Nuclear Instruments and Methods in Physics Research Section A: Accelerators, Spectrometers, Detectors and Associated Equipment* 1066 (2024), p. 169646. ISSN: 0168-9002. DOI: https://doi.org/10.1016/j.nima.2024.169646. URL: https://www.sciencedirect.com/science/article/pii/S0168900224005722.

[23] T. Zolkin, S. Nagaitsev, I. Morozov, S. Kladov, and Y.-K. Kim. *Isochronous and period-doubling diagrams for symplectic maps of the plane*. 2024. arXiv: 2412.05541 [nlin.CD]. URL: https://arxiv.org/abs/2412.05541.

[24] T. Zolkin, S. Nagaitsev, I. Morozov, S. Kladov, and Y.-K. Kim. *Dynamics of McMillan mappings III. Symmetric map with mixed nonlinearity*. 2024. arXiv: 2410.10380 [nlin.SI]. URL: https://arxiv.org/abs/2410.10380.

[25] T. Zolkin, S. Nagaitsev, and I. Morozov. *Dynamics of McMillan mappings I. McMillan multipoles*. 2024. arXiv: 2405.05652 [nlin.SI]. URL: https://arxiv.org/abs/2405.05652.

[26] V. Gauzshtein et al. "Measurement of the tensor analyzing power T20 for incoherent photoproduction on a deuteron above the first resonance region". In: *Physics Letters B* 860 (2025), p. 139166. ISSN: 0370-2693. DOI: https://doi.org/10.1016/j.physletb.2024.139166. URL: https://www.sciencedirect.com/science/article/pii/S037026932400724X.

[27] Y. V. Shestakov et al. "Photodisintegration of tensor polarized deuteron: Status of the experiment at the VEPP-3 storage ring". In: *International Journal of Modern Physics E* 0.0 (0), p. 2441023. DOI: 10.1142/S0218301324410234. eprint: https://doi.org/10.1142/S0218301324410234. URL: https://doi.org/10.1142/S0218301324410234.

[28] T. Zolkin, I. Morozov, and S. Nagaitsev. "Understanding sextupole". English. In: *Proc. 15th International Particle Accelerator Conference* (Nashville, TN). IPAC'24 - 15th International Particle Accelerator Conference 15. JACoW Publishing, Geneva, Switzerland, May 2024, pp. 725–728. ISBN: 978-3-95450-247-9. DOI: 10.18429/JACoW-IPAC2024-MOPS13. URL: https://indico.jacow.org/event/63/contributions/3720.

[29] I. Morozov and Y. Maltseva. *Coupled Twiss Parameters Estimation from TbT Data*. 2023. arXiv: 2311.14287 [physics.acc-ph].

[30] I. Morozov and Y. Maltseva. *Combining Methods for Localization of Linear Focusing Errors*. 2023. arXiv: 2311.10570 [physics.acc-ph].

[31] V. V. Anashin et al. "Experiments with the KEDR Detector at the e^+e^- Collider VEPP-4M in the Energy Range $\sqrt{s}$ = 1.84–3.88 GeV". In: *Physics of Particles and Nuclei* 54.1 (Feb. 2023), pp. 185–226. DOI: 10.1134/s1063779623010033.

[32] V. V. Anashin et al. "Measurement of J/ψdecays into final states $2(\pi^+\pi^-)\pi^0$, $K^+K^-\pi^+\pi^-\pi^0$, $2(\pi^+\pi^-)$ and $K^+K^-\pi^+\pi^-$". In: *The European Physical Journal C* 82.10 (Oct. 2022). DOI: 10.1140/epjc/s10052-022-10879-9.

[33] T. Zolkin, S. Nagaitsev, and I. Morozov. "McMillan map and nonlinear Twiss parameters". In: (2022). DOI: 10.48550/ARXIV.2204.12691.

[34] V. V. Anashin et al. "Measurement of the branching fraction of J/ → at KEDR". In: (2022). DOI: 10.48550/ARXIV.2211.13520.

[35] G. Baranov, A. Bogomyagkov, I. Morozov, S. Sinyatkin, and E. Levichev. "Lattice optimization of a fourth-generation synchrotron radiation light source in Novosibirsk". In: *Physical Review Accelerators and Beams* 24.12 (Dec. 2021). DOI: 10.1103/physrevaccelbeams.24.120704.

[36] I. Morozov and P. Piminov. "Detection of Anomalies in BPM Signals at the VEPP-4M". en. In: *Proceedings of the 27th Russian Particle Accelerator Conference* RuPAC2021 (2021), Russia. DOI: 10.18429/JACOW-RUPAC2021-WEPSC40.

[37] I. Morozov, P. Piminov, and I. Yakimov. "Precise Analysis of Beam Optics at the VEPP-4M by Turn-by-Turn Betatron Phase Advance Measurement". en. In: *Proceedings of the 27th Russian Particle Accelerator Conference* RuPAC2021 (2021), Russia. DOI: 10.18429/JACOW-RUPAC2021-WED04.

[38] P. Piminov et al. "VEPP-4M Collider Operation at High Energy". en. In: *Proceedings of the 12th International Particle Accelerator Conference* IPAC2021 (2021), Brazil. DOI: 10.18429/JACOW-IPAC2021-MOPAB034.

[39] P. Piminov et al. "VEPP-4M Electron Positron Collider Operation at High Energy". en. In: *Proceedings of the 27th Russian Particle Accelerator Conference* RuPAC2021 (2021), Russia. DOI: 10.18429/JACOW-RUPAC2021-TUA01.

[40] V. Anashin et al. "Precise measurement of Ruds and R between 1.84 and 3.72 GeV at the KEDR detector". In: *Physics Letters, Section B: Nuclear, Elementary Particle and High-Energy Physics* 788 (2019), pp. 42–51. DOI: 10.1016/j.physletb.2018.11.012.

[41] V. Anashin et al. "Measurement of Γee x Bμμ for (2S) meson". In: *arXiv* (2018).

[42] V. Anashin et al. "Measurement of Γee x Bμμ for (2S) meson". In: *Physics Letters, Section B: Nuclear, Elementary Particle and High-Energy Physics* 781 (2018), pp. 174–181. DOI: https://doi.org/10.1016/j.physletb.2018.03.079.

[43] V. Anashin et al. "Precise measurement of Ruds and R between 1.84 and 3.72 GeV at the KEDR detector". In: *arXiv* (2018).

[44] I. Morozov and E. Levichev. "Dynamical Aperture Control in Accelerator Lattices With Multipole Potentials". en. In: *CERN Proceedings* (2017), Vol 1 (2017): CERN–BINP Workshop for Young Scientists in e+e. DOI: 10.23727/CERN-PROCEEDINGS-2017-001.195.

[45] I. Morozov and E. Levichev. "Dynamical Aperture Control in Accelerator Lattices With Multipole Potentials". In: *IPAC 2016 - Proceedings of the 7th International Particle Accelerator Conference* (2016), pp. 3455–3457.

[46] Y. Maltseva and I. Morozov. "Model-dependent accelerator lattice fit based on BPM data and generating functions". In: *IPAC 2016 - Proceedings of the 7th International Particle Accelerator Conference* (2016), pp. 3311–3313.

[47] T. Zolkin, Y. Kharkov, I. Morozov, and S. Nagaitsev. "A Model Ring With Exactly Solvable Nonlinear Motion". In: (Sept. 2013).

[48] G. Stancari et al. "Beam halo dynamics and control with hollow electron beams". In: *HB 2012 - Proceedings of the 52nd ICFA Advanced Beam Dynamics Workshop on High-Intensity and High-Brightness Hadron Beams* (2013), pp. 466–470.

[49] T. Zolkin, Y. Kharkov, I. Morozov, and S. Nagaitsev. "Nonlinear Accelerator with Transverse Motion Integrable in Normalized Polar Coordinates". In: *IPAC 2012 - International Particle Accelerator Conference 2012* (2012), pp. 1116–1118.

[50] I. Morozov, G. Stancari, A. Valishev, and D. Shatilov. "Simulation of Hollow Electron Beam Collimation in the Fermilab Tevatron Collider". In: *IPAC 2012 - International Particle Accelerator Conference 2012* (2012), pp. 94–96.

[51] i-a-morozov. *i-a-morozov/matrix: v0.1.0.* Version v0.1.0. Feb. 2026. DOI: 10.5281/zenodo.18787029. URL: https://doi.org/10.5281/zenodo.18787029.

[52] i-a-morozov. *i-a-morozov/domain: v0.1.0.* Version v0.1.0. Feb. 2026. DOI: 10.5281/zenodo.18787033. URL: https://doi.org/10.5281/zenodo.18787033.

[53] I. Morozov. *i-a-morozov/harmonica: v0.1.2.* Version v0.1.2. Feb. 2026. DOI: 10.5281/zenodo.18786177. URL: https://doi.org/10.5281/zenodo.18786177.

[54] I. Morozov. *i-a-morozov/ppt: v0.1.0*. Version v0.1.0. July 2025. DOI: 10.5281/zenodo.16408965. URL: https://doi.org/10.5281/zenodo.16408965.

[55] I. Morozov. *i-a-morozov/elementary: v0.1.0*. Version v0.1.0. June 2025. DOI: 10.5281/zenodo.15771385. URL: https://doi.org/10.5281/zenodo.15771385.

[56] I. Morozov. *i-a-morozov/sympint: v0.2.0*. Version v0.2.0. Apr. 2025. DOI: 10.5281/zenodo.15265930. URL: https://doi.org/10.5281/zenodo.15265930.

[57] I. Morozov. *i-a-morozov/model: v0.2.0*. Version v0.2.0. Oct. 2024. DOI: 10.5281/zenodo.13933324. URL: https://doi.org/10.5281/zenodo.13933324.

[58] I. Morozov. *i-a-morozov/tohubohu: v0.1.1*. Version v0.1.1. Apr. 2025. DOI: 10.5281/zenodo.15265932. URL: https://doi.org/10.5281/zenodo.15265932.

[59] i-a-morozov. *i-a-morozov/square: v0.1.0*. Version v0.1.0. Nov. 2024. DOI: 10.5281/zenodo.14064955. URL: https://doi.org/10.5281/zenodo.14064955.

[60] I. Morozov. *i-a-morozov/twiss: v0.2.6*. Version v0.2.6. Aug. 2024. DOI: 10.5281/zenodo.13327959. URL: https://doi.org/10.5281/zenodo.13327959.

[61] I. Morozov. *i-a-morozov/ndmap: v0.1.0*. Version v0.1.0. Nov. 2023. DOI: 10.5281/zenodo.10158382. URL: https://doi.org/10.5281/zenodo.10158382.

[62] I. Morozov. *i-a-morozov/SIGNAL: 2021 release update*. Version v1.2.0. May 2021. DOI: 10.5281/zenodo.4741546. URL: https://doi.org/10.5281/zenodo.4741546.

[63] I. Morozov. *i-a-morozov/PROPAGANDA v1.0.1*. Version v1.0.1. Oct. 2020. DOI: 10.5281/zenodo.4075909. URL: https://doi.org/10.5281/zenodo.4075909.